

Clinical Approach to the Male Patient

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Learning Objectives

- Recognize common and concerning symptoms in male patients and generate a list of differential diagnoses for such symptoms.
- Describe techniques for conducting male genital and digital rectal examinations and interpret examination findings.
- Differentiate between penile abnormalities (epispadias, hypospadias, phimosis, Peyronie's disease, condyloma, Bowen disease, invasive carcinoma) in clinical scenarios.
- Identify and distinguish scrotal/testicular abnormalities (hydrocele, hematocele, spermatocele, varicocele, orchitis, epididymitis, testicular tumor) in clinical vignettes.
- Differentiate between prostate abnormalities (prostatitis, benign prostatic hypertrophy, prostate cancer) in clinical cases.
- Differentiate between hernias (direct and indirect inguinal hernia, femoral hernia) in clinical vignettes.
- Describe the clinical evaluation and management approaches for acute scrotum, scrotal mass, and male sexual dysfunction.

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The learning objectives above require that you synthesize clinical presentation with pathology and anatomy content.

Your board questions on this content will be very integrated, but the first step will inevitably be recognizing a patient problem (diagnosis) from a description in a clinical vignette.

This lecture attempts to provide a bit more detail about how patients present in the physician office or ED when they experience problems related to their reproductive organs, in hopes of helping you move beyond simple buzzword associations.

Common Male Presenting Concerns/Complaints

- Changes in sexual function and response
 - Erectile dysfunction
 - Loss of libido
 - Ejaculatory or orgasmic dysfunction
- Penile discharge or lesions
- Scrotal pain, swelling or masses
- Urinary symptoms
 - Dysuria
 - Hematuria
 - Hesitancy
 - Reduced stream
 - Double voiding, terminal dribbling
- Other
 - Hernia
 - Blood in ejaculate

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These are the concerns commonly expressed by male patients when they are having problems with their reproductive system. After you have learned the pathology of the male reproductive system, you should be able to look at each complaint and generate a differential diagnosis list of the possible diseases/disorders that could cause these symptoms.

Medical History and Health Promotion

- Medical History (in addition to routine medical history elements):
 - <https://www.cdc.gov/std/treatment/sexualhistory.pdf>
 - 5 Ps: **P**artners, **P**ractices, **P**rotection from STDs, **P**ast history of STDs, **P**revention of Pregnancy
 - Minimum: elicit sexual orientation, number of partners in last month and history of STIs
 - ROS symptoms to cover in HPI: urinary symptoms; penile discharge, pain or lesions; scrotal pain, swelling or masses; sexual dysfunction etc.
- Health Promotion:
 - Condom use
 - Reducing risk of STDs (STIs) and HIV by limiting number of partners
 - STD screening
 - Prompt attention for genital lesions or penile discharge
 - Testicular self-examination (TSE)

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The absolute essentials to a male medical history and potential topics for patient education are listed above.

Think for a moment. Though you must cover the above elements in a routine screening health physical exam for a male patient and you might cover these elements in office visits for other presenting problems, you would not necessarily cover these items in an ED visit for acute substernal crushing chest pain!

Just like every aspect of clinical medicine, clinical judgment must be applied. However, don't skip these just because you are not comfortable talking about them!

Sexual Function: How to Ask

- Non-judgmental stance is most important
- Prepare the patient: *"I am going to ask you a few questions about your sexual health and sexual practices. I understand that these questions are very personal, but they are important for your overall health. Just so you know, I ask these questions to all of my adult patients. Like the rest of our visits, this information is kept in strict confidence. Do you have any questions before we get started?"*
- "Is your sex life satisfying?"
 - Any problems with desire?
 - Any concerns about performance, frequency, duration of intercourse?
 - Any problems with achieving or maintaining an erection?
 - Any problems with ejaculation?

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Think about how you communicate "non-judgment" to another person, both verbally and non-verbally.

How comfortable YOU are with the topic will often dictate how comfortable the patient is with the topic. Don't assume that people don't want to talk about their sex life. Many patients crave an open opportunity to talk, where they feel safe and confident about the quality of the information they are receiving.

If the patient has concerns about erectile dysfunction:

- “Do you have problems achieving an erection, maintaining an erection or both?”
- “Do you have morning erections?”
- “Describe your relationship(s).”
 - “Is your relationship satisfying?”
 - “Are there any communication problems or conflicts right now?”
- “Describe your stress level.”
- Screen for depression (SIGECAPS)
- Physician must specifically also ask about medication use, risk factors and symptoms of systemic disease: diabetes, cardiovascular disease, endocrine disorder

Do not assume ANY erectile dysfunction is part of normal aging!

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Here are the questions and topics that need to be covered in the medical history of a patient presenting with erectile dysfunction.

A detailed sexual history will inevitably delve into psychological issues and stressors. Again, don't assume that people don't want to talk about this. Ask, and if the patient does not want to talk about it, they will let you know. Further, if the information is important to your clinical management, you need to press at least a bit:

“I understand that you don't want to talk about this. First, anything you tell me is completely confidential; and second, I asked because the answer to this question will help me determine what is wrong and how I can best help you. I want the very best for you and I want to help.”

Male Genital Exam – Techniques

**National Clinical Training Center for Family Planning
(NCTCFP)**

<https://www.ctcfp.org/male-genital-exam/>

Bates' Visual Guide

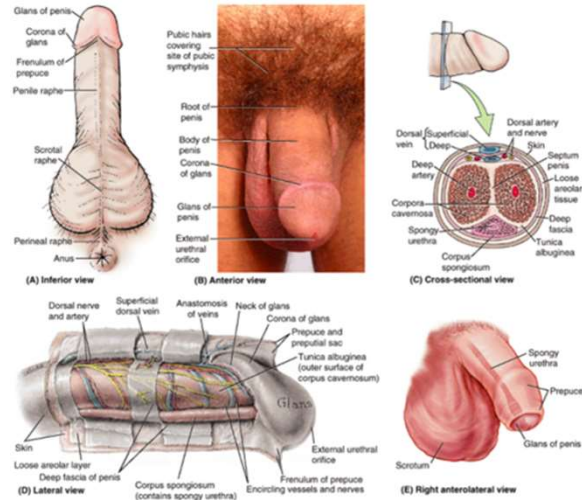
Male Genitalia, Rectum, Anus and Prostate

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These video presentations by the NCTCFP and Bates' are rich with detail on the male genital, hernia and rectal examinations.

Male Genital Examination - Techniques

- Genital exam may be done with patient standing or supine; **hernia examination must be done with patient standing**
- Provide orienting statements
- If patient develops an erection, explain that this is a natural response and proceed with an unruffled demeanor
- Wear gloves



Review anatomy!

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After collecting a medical history, you will perform a physical examination and this should include a male genital examination, and in some instances, a hernia and rectal examination. We will briefly review male genital rectal and hernia exam techniques here and you can go to your Bates reference book for additional detail. However, practice makes perfect, and you will have opportunities to perform these examinations on live patients during your clerkship experiences. Right now, you can reflect on the words you will use to orient the patient to the procedures.

Male Genital Examination – What to Say

“Today I will be examining your genitals. This will involve looking closely for any abnormalities and palpating, which is touching, some of the structures to check for lumps or irregularities. The examination is necessary to screen for any problems that you may not have noticed yourself (or to determine the cause of your symptoms). I will be gentle and none of this should hurt, so please let me know if you feel any discomfort.

After the genital examination, I will be performing a digital rectal examination. I will be inserting a gloved, lubricated finger into your rectum to check your prostate. This will feel a bit like a bowel movement and should not hurt. If you experience any discomfort, again, please let me know.

For both examinations, I will be letting you know what I am doing as I go along so there will be no surprises.”

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Many patients will have already had an examination performed in the past. Patients who have not will require extra sensitivity and support.

Male Genital Examination

Penis

- **Inspection**: look closely at the entire penis, dorsal and ventral surfaces and urethral meatus. Have the patient retract the foreskin, if present. Note any difficulties.
- **Palpation**: gently compress the glans to inspect the urethral meatus for inflammation/discharge. Palpate the shaft of the penis for any induration. Palpate any visible abnormality and note induration or tenderness.



Inspection of urethral meatus. Any discharge should be tested.

Urethral discharge, either spontaneous or elicited during exam, should raise suspicion for an STI, usually chlamydia or gonorrhea.

If you see a discharge, be prepared to obtain a sample for culture or PCR test.

Male Genital Examination

- **Scrotum and Contents:**

- **Inspection:** inspect the skin, lifting the scrotum to inspect posterior surface. Inspect pubic hair and distribution. Note scrotal contours and symmetry. Inspect inguinal areas for erythema, excoriation, visible adenopathy.
- **Palpation:** using two hands, cradle the testis at both upper and lower poles in the thumb and fingertips of both hands. Palpate the scrotal contents, gently sliding them back and forth from the fingertips of one hand to the other.
- Assess size, shape, consistency of each testis
- Note tenderness
- Palpate (do not squeeze!) the epididymis on the posterior surface of each testis; note tenderness
- Palpate each spermatic cord between thumb and fingers
- Palpate any nodules or swellings

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If the patient complains of any scrotal pain or swelling, this exam becomes imperative. However, it also must be done very gently.



Palpating left testicle



Palpating spermatic cord

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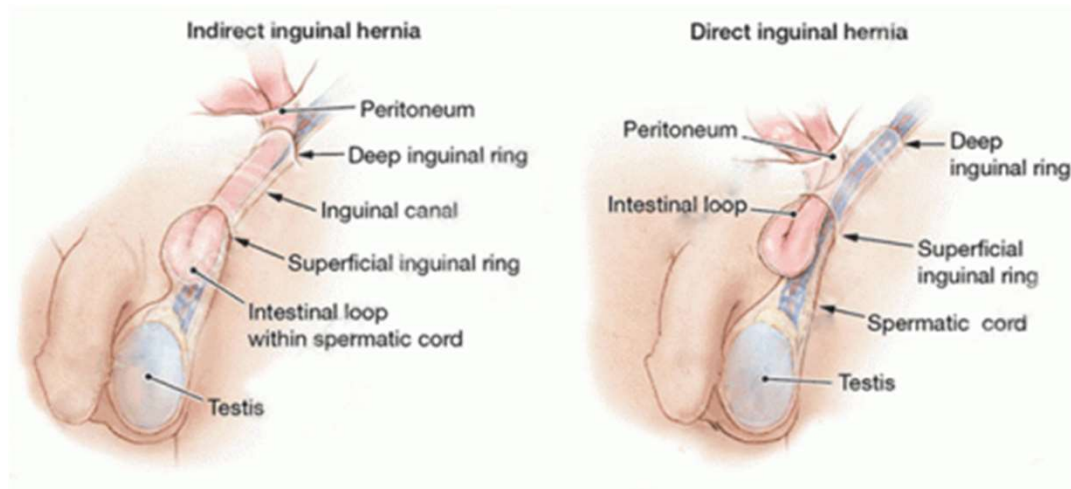
These are images from the Bates textbook.

The most important message in palpating scrotal contents is to be gentle but be thorough and deliberately identify all contents by palpation bilaterally!

Also Part of the Male Examination

- Hernia
 - Inspection
 - Palpation
- Digital Rectal Exam

Hernias



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This is a review of the anatomy of direct and indirect hernias. Make sure you know the difference anatomically.

Remember, **Indirect inguinal hernias** are the most common type of hernia in males and females.

Hernia Examination - Technique

- **Inspection:** with patient standing inspect the inguinal regions and genitalia for bulging areas and asymmetry
- **Palpation:**
 - Place tip of dominant index finger at the anterior inferior margin of the scrotum and move the finger and hand upward toward the superficial inguinal ring, invaginating the scrotal skin beneath the peripubic fat pad next to the base of the penis
 - Follow the spermatic cord upward to the superficial inguinal ring
 - Ask the patient to cough and simultaneously note any bulge that taps against your fingertip
- Examine both sides in this manner



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Please keep your fingernails as short as possible if you will be touching patients!

An inguinal hernia will be apparent if you feel a bulge or tap against your fingertip as it sits in the inguinal canal at the deep (internal) inguinal ring next to the pubic tubercle (technique shown above) when THE PATIENT COUGHS OR BEARS DOWN WITH THE VALSALVA MANEUVER.

Hernia symptoms: groin heaviness or pain - burning or aching; symptoms worse at the end of the day or with heavy physical activity; a bulge in the groin or mass in the scrotum that may reduce in size when the patient is supine.

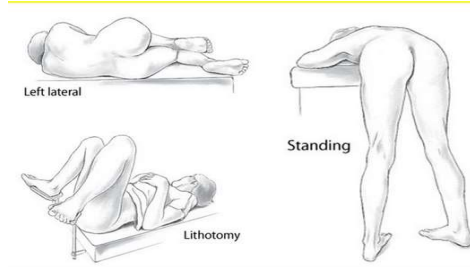
Digital Rectal Examination

Patients (not performed on every male patient!):

- Men > 50 years old
- Men > 40 years old who have a positive family hx (prostate or colon Ca)
- Men with symptoms at any age (prostate sx or rectal bleeding)

Position of patient

- Standing
- Left lateral decubitus



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Note the various positions that work for the rectal examination. Standing and left lateral decubitus positions are most commonly used. This exam is often performed in appropriate patients after the male genital examination, thus it is easy to transition from one to the other when both are done with the patient in the standing position.

The lithotomy position:

The term lithotomy means “stone removal” as this is the position that allows easiest access to the urethra and bladder (and the vagina and uterus in female patients). We don’t usually put male patients into this position except when they are undergoing urologic procedures.

Digital Rectal Examination - Technique

- Provide orienting statements
- Wear gloves and lubricate index finger
- **Inspection:** perianal and sacrococcygeal lumps, ulcers, inflammation, rashes or excoriations
- Instruct patient to bear down
- Place pad of index finger over anus
- Gently insert fingertip into anal canal, pointing toward umbilicus
- Proceed to advance finger as sphincter relaxes
 - If tenderness is present, halt and look for lesions such as anal fissure
 - Rotate hand clockwise and counterclockwise to assess rectal surface
 - Palpate the posterior surface of the prostate gland on the anterior wall of the rectum: identify lateral lobes and median sulcus, size, shape, consistency, nodules, tenderness.
- Withdraw finger and perform hemoccult test on any fecal matter or mucus on the glove.

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Penile Problems-Differential Diagnosis

- Lesions

- Genital warts
- Genital herpes
- Syphilitic chancre
- Bowen's disease
- Squamous cell carcinoma
- Balanitis (reactive arthritis)
- Excoriations – pubic lice, scabies



Pearly penile papules

- Discharge: **Chlamydia and Gonococcal urethritis, Reactive arthritis**
- Urethral meatus location: **Hypospadias, Epispadias**
- Prepuce retraction problem: **Phimosis/ Paraphimosis**
- Palpable non-tender hard plaque, angulated penis: **Peyronie's disease**
- Palpable ventral surface induration/ tenderness: **urethral stricture or possible carcinoma**

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This slide focuses briefly on clinical presentations and differential/provisional diagnoses. Patients may present with no complaints or symptoms with some of these diseases. Other patients will present with pain, discomfort with urination or lesions or perhaps a sexual concern (eg, pain or difficulty with erections or intercourse can occur with Peyronie's disease). Thus, the history AND physical examination are essential.

Phimosis

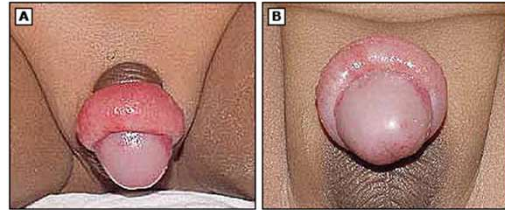
- **Inability to retract the foreskin.**
- May be normal in children and typically resolves by age 5.
- Leads to hygiene and cleaning problems
- Treatment is required only if there are any complications such as balanitis, UTI, or urinary obstruction.
- Rx:
 - Betamethasone cream
 - Gentle and gradual stretching of foreskin
 - Circumcision – preferred surgical option



Paraphimosis

- Entrapment of the foreskin in the retracted position.
- **It is a medical emergency.**
- Retraction may occur during catheterization or physical examination.
- Retracted foreskin functions as a tourniquet, causing glans to swell → worsening the constriction → vascular compromise and necrosis
- Rx:
 - Firm circumferential pressure on glans
 - Dorsal slit under local anesthesia
 - Finally, circumcision to prevent recurrence

Paraphimosis



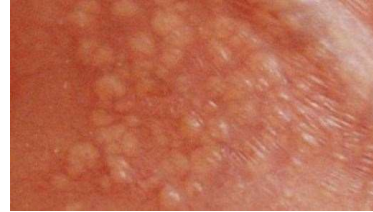
Paraphimosis caused by excessive retraction of a physiologic phimosis in a male infant. Note the marked swelling at the coronal sulcus and the flaccid penile shaft.

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UpToDate®

Scrotal Problems – Differential Diagnosis

- Visible excoriations – pubic lice
- Skin lesions:
 - Small scar of vasectomy
 - Superficial epidermoid cysts, folliculitis
 - Tinea cruris – dermatophyte infection
 - Molluscum contagiosum (translucent, umbilicated, dome-shaped papules)
- Scrotal edema – generalized edema of CHF or nephrotic syndrome
- Scrotal mass/swelling
 - Nontender, fluid-filled, transilluminates: **Hydrocele**
 - Nontender soft mass, fingers cannot get above mass within scrotum: **Indirect inguinal hernia**
 - Swelling occurring after blunt trauma: **consider Hematocele**



Fordyce spots – enlarged sebaceous glands

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Pubic lice may be associated with regional itching.

Patients may note epidermoid cysts and worry that they are a disease!

Folliculitis can be painful or uncomfortable.

Tinea cruris causes rash/redness and often itching/burning.

Molluscum contagiosum does not hurt, but the appearance alarms patients.

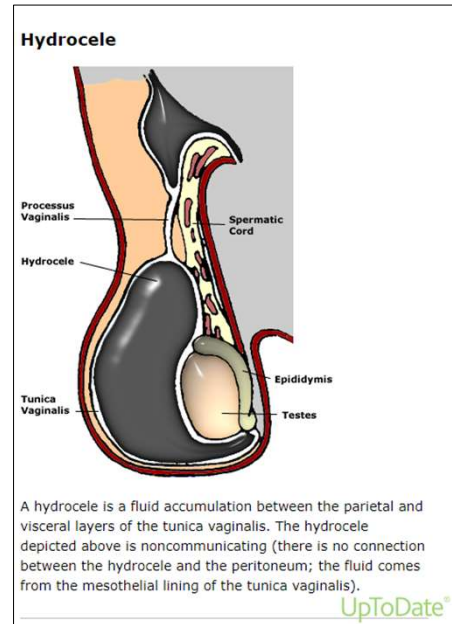
Inguinal Hernia

- **Protrusion of the abdominal contents through an acquired or congenital area of weakness or defect in the abdominal wall just above the inguinal ligament.**
 - **Indirect:** Traversing the deep (internal) inguinal ring into the inguinal canal
 - **Direct:** Extending directly forward and not passing through the inguinal canal
- Uncomplicated hernias are typically reducible. (unlike other scrotal swellings)
- **Examining fingers cannot get above the swelling.** (unlike hydrocele and other localized swellings)
- Most are asymptomatic, but some become incarcerated (cannot be reduced) or strangulated (compromised blood supply), causing pain.
- Diagnosis is clinical.
- Treatment is elective surgical repair.



Hydrocele

- **Collection of peritoneal fluid between the parietal and visceral layers of the tunica vaginalis, which directly surrounds testis and spermatic cord.**
- Non-tender, fluid-filled mass; size ranges from small, soft collections to massive, tense collections of several liters
- **Hydrocele fluid transilluminates well**, which differentiates the process from a possible hematocele, hernia, or solid mass.
- Treatment is indicated in patients with symptoms
- Common surgical procedure is excision of the hydrocele sac.



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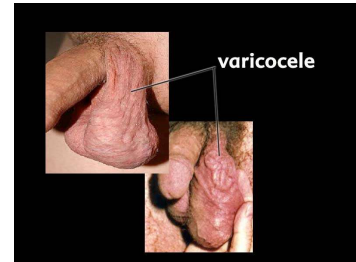
Recall embryology: Incomplete obliteration of processus vaginalis may lead to congenital hydrocele in infants.

Hematocele

- **A hematocele, a blood-filled scrotum, may occur when there is damage to scrotal contents.**
- Scrotal swelling and ecchymosis can occur after scrotal trauma.
- Look of tender swelling and risk factors (eg, trauma, surgery, bleeding disorder or use of anticoagulants)
- The swelling does not transilluminate.
- USG may be helpful

Testicular, Epididymis and Spermatic Cord Problems – Differential Diagnosis

- Absent or atrophic testis: **Cryptorchidism** (increases risk of testicular Ca)
- Small testes (normal length = 3.5 cm)
 - Klinefelter syndrome (usually ≤ 2 cm)
 - Soft, atrophic testes: cirrhosis, myotonic dystrophy, estrogen use, hypopituitarism, or following orchitis
- Tender, inflamed, swollen testis
 - Orchitis (mumps)
 - Acute epididymitis
 - Torsion of the spermatic cord
- Painless mass/nodule
 - Testicular cancer
 - Spermatocele
 - Epididymal cyst
- “Bag of worms” = Varicocele (varicose veins of the spermatic cord)



These transilluminate

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Most men don't know if they have small testes.

Orchitis, epididymitis and torsion all hurt, usually significantly! The scrotum may be erythematous and swollen. Both epididymitis and orchitis can be associated with fever.

A **spermatocele** is palpable as a nontender mass somewhere along the spermatic cord.

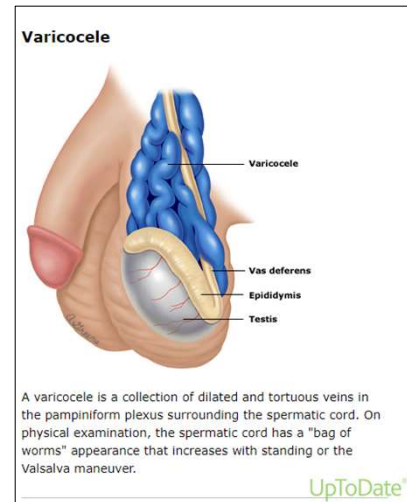
Similarly an **epididymal cyst** will be palpable along the epididymis. If you cannot confidently identify these as separate mass from the testicle during physical examination, then you may have to further evaluate for testicular tumor.

Varicoceles don't hurt, but are a source of concern for patients.

Thus, you cannot skip the male genital examination when the patient has no complaints.

Varicocele

- **Dilatation of the pampiniform plexus of spermatic veins.**
- Present in 15-20% of post-pubertal males
- Generally left-sided, first appears at puberty, and may become larger over time. Why left sided?
- PE: minimal left-sided scrotal fullness on Valsalva to a large, soft, left-sided scrotal mass ("**bag of worms**") that decompresses and disappears in the recumbent position.
- May be associated with dull pain, testicular atrophy (due to high temp), and decreased fertility
- Rx: surgical ligation or percutaneous venous embolization



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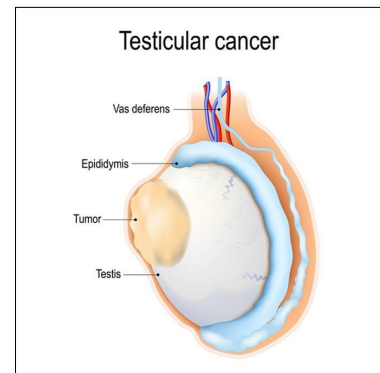
Varicoceles are generally left sided because the left gonadal vein drains into the left renal vein, which crosses over to the inferior vena cava and can be compressed between the aorta and the superior mesenteric artery. The **left renal vein** can be compressed by the superior mesenteric artery or abdominal masses (lymphadenopathy). Compression of the left renal vein by the superior mesenteric artery occurs more frequently in subjects with little intra-abdominal adipose tissue (anorexia). This is also sometimes referred to as 'nutcracker effect.' The pampiniform plexus is the site of dilation in varicoceles, but the obstruction is usually in the renal vein.

Varicoceles are rarely seen on the right side because the right renal vein does not cross over the aorta; therefore, it is unlikely to be compressed and obstructed.

Attention: Left-sided varicocele that do not resolve in supine position should raise suspicion for a tumor such as **renal cell carcinoma** that obstructs the inferior vena cava or left renal vein.

Testicular Cancer

- **Begins as a painless scrotal mass**
- Most common solid cancer in males of age 15-35
- Most common testicular cancer → **seminoma**
- Symptoms: painless scrotal mass, testicular heaviness
- Diagnosis: USG
- Treatment: radical orchiectomy with radiation, chemotherapy or combination



Any nodule within the testis warrants investigation for malignancy!

Prostate Problems – Differential Diagnosis

- Normal prostate is about 2.5 cm long, heart shaped, rubbery and non-tender
- Prostate size on DRE is often reported in “fingerbreadths”; 2 or less fingerbreadths is normal size
- Differential diagnosis based on symptoms and signs:
 - Urinary sx (hesitancy, double voiding, decreased stream, terminal dribbling) – **BPH, Prostate cancer, Prostatitis**
 - Dysuria, back pain, pelvic pain, ejaculatory pain – **Prostatitis**
 - Tender, boggy, nodular, hot prostate – **Prostatitis**
 - Uniformly enlarged non-tender prostate – **BPH or cancer**
 - Indurated, non-tender nodular or irregular prostate – **Cancer**

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Especially for men over 50, the above urinary symptoms are important. If the patient does not volunteer them, you need to ask, given the high prevalence of BPH in older men.

BPH and Prostate Cancer

- BPH is nonmalignant adenomatous overgrowth of the periurethral prostate gland.
- Prostate Ca is usually adenocarcinoma.
- Symptoms of bladder outlet obstruction – weak stream, hesitancy, urinary frequency, urgency, nocturia, incomplete emptying, terminal dribbling, incontinence, and complete urinary retention.
- DRE:
 - BPH: enlarged and nontender prostate, rubbery consistency, loss of the median furrow
 - Prostate Ca: Firm, hard, irregular area in the prostate

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Evaluation and Management of Prostate Problems

- Start with a thorough H & P
- Most symptomatic patients will require a referral to a urologist for evaluation and treatment options
- **Routine PSA testing for men is not recommended by USPSTF (2012)**
- These guidelines are being questioned and debated
- What if you discover or suspect a prostate mass/cancer on DRE?
 - Assess PSA
 - Refer for biopsy
 - CT scan is used to assist with staging of prostate cancers

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Hemospermia (blood in ejaculate)

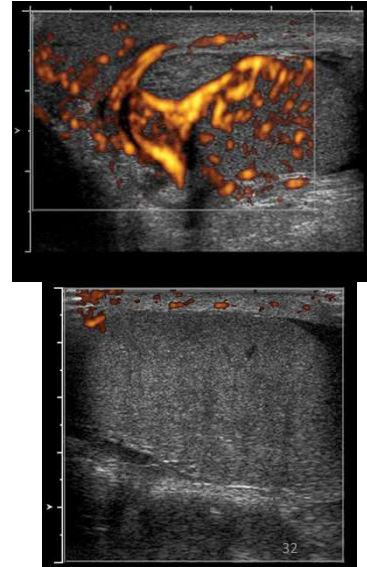
- Believed to be common, but true prevalence is unknown
- Of great concern when men discover it
- Most common cause - idiopathic
- Most common known cause - recent urogenital surgery
- Rarely associated with malignancy
- Age < 40, urogenital infections are common (urethritis, prostatitis, epididymitis, inflammation of seminal vesicles)
- Age > 40, rule out malignancy

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Semen is composed of sperm from the distal epididymis and fluids from the seminal vesicles, prostate, and Cowper and bulbourethral glands. Thus, a lesion anywhere along this pathway could introduce blood into the semen.

Evaluation and Management of “Acute Scrotum”

- Acute scrotal pain and swelling DDX: **Testicular torsion, Epididymitis, Torsion of testicular appendix**
- **Testicular torsion is a surgical emergency!**
- How to tell the difference?
 - History and PE are poor differentiators:
 - Nausea/ vomiting, high position of the testicle on examination, absent cremasteric reflex → **Testicular torsion**
 - Dysuria, fever, scrotal erythema → **Epididymitis**
 - Nodule and “blue dot sign” → **Torsion of testicular appendix**
 - Ultrasound is sensitive and specific, but only if the technical expertise of the user is high!
 - Get immediate urologic consultation



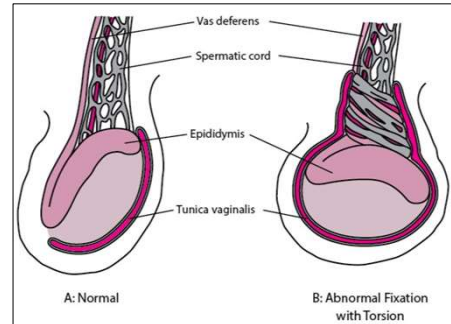
Acute onset of scrotal/testicular pain is a **medical emergency!**

Note that surgical exploration is often necessary for definitive diagnosis because it is **IMPERATIVE** that we don't miss testicular torsion. The testicle will become ischemic and will “die” if the torsion is not quickly reduced. Missing this diagnosis is a medical liability. For primary care physicians, **KNOW** that this can indicate testicular torsion and refer the patient for urological consultation **IMMEDIATELY!** Thus, when you communicate with the urologist or his/her office, you **MUST** communicate your concern about the diagnosis so that the condition is addressed immediately.

Even though manual detorsion can be done and you may learn this procedure in clerkship or residency, it is standard of care for all patients to have surgery to “fixate” the testicle in the scrotum so that this does not happen recurrently!

Testicular Torsion

- **An emergency condition** due to rotation of the testis and consequent strangulation of its blood supply.
- **Symptoms:** acute scrotal pain and swelling, nausea, and vomiting
- **Diagnosis:** based on PE and confirmed by color Doppler USG.
- **PE:** tender and erythematous testis, testis may be elevated (high-riding) and horizontal, absent cremasteric reflex
- **Treatment:** immediate manual detorsion followed by surgical intervention.

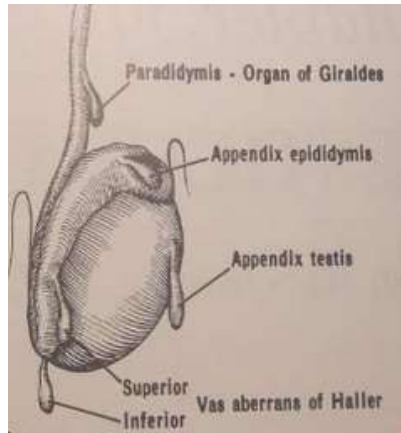


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- Poor fixation of testis to tunica vaginalis leads to increased risk of twisting of spermatic cord leading to testicular torsion.
- Sudden onset of severe, unilateral, constant testicular or inguinal pain with nausea, vomiting. Physical exam: swelling and redness; elevated (high-riding)/horizontally placed testis; absent cremasteric reflex. No or diminished blood supply to testis on color doppler ultrasound. → suggest testicular torsion. Consider urgent surgical detorsion
- What is cremasteric reflex?: Stroking the medial (inner) side of thigh leads to elevation of testis on same side.
- Testicular torsion may lead to venous congestion, hemorrhagic infarction and necrosis of affected testis.

Torsion of Testicular Appendix

Blue Dot Sign



**Testicular appendages may be present in several locations.
An appendage is vulnerable to torsion.**

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The appendix testis is a 0.3 cm pedunculated structure on the anterosuperior aspect of the testis. Torsion can produce testicular pain that ranges from mild to severe. Infarction and necrosis of the appendix testis can be seen as a "blue dot" sign in 21% of cases. (Ref: UpToDate)

Epididymitis

- Inflammation of the epididymis, occasionally accompanied by inflammation of the testis (epididymo-orchitis).
- Scrotal pain and swelling usually occur unilaterally. Also, fever, nausea and urinary symptoms.
- Diagnosis is based on physical examination.
- **PE: swelling, redness, induration, marked tenderness over posterior side of testis. Prehn sign (manual elevation of scrotum relieves the pain) positive.**
- Treatment is with antibiotics, analgesics, and scrotal support.



Epididymo-orchitis can result in painful, swollen, erythematous testes and epididymis, as shown in this photo.

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- Scrotal pain with urinary symptoms (such as dysuria, increased frequency) and fever. Physical exam: swelling, redness, induration and tenderness over posterior of testis. Prehn sign (manual elevation of scrotum relieves the pain) positive. Intact cremasteric reflex. Urinalysis may show WBCs. → suggest epididymitis
- Epididymitis in young male is usually due to STI. Consider testing for *Neisseria gonorrhoeae* and *Chlamydia trachomatis*.
- Cremasteric reflex remains intact in epididymitis. (unlike in testicular torsion)

Evaluation and Management of Scrotal Mass

Hydrocele vs. Hernia vs. Tumor

- If a scrotal mass is detected:
 - Tender or non-tender?
 - Can you get your fingers above the mass in the scrotum? (if Yes? Not a hernia!)
 - Listen to the mass with a stethoscope. Are bowel sounds present? (if Yes? Hernia is likely!)
 - **Transillumination**: in a darkened room, use a strong flashlight or penlight to backlight the scrotum through the mass
 - Transilluminates (light gets through), likely fluid filled: Hydrocele, Spermatocele
 - Does not transilluminate, solid tumor more likely



This scrotal mass transilluminates.
This is a hydrocele.

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Many scrotal conditions and masses are nontender but this does not preclude pursuing a diagnosis.

Evaluation and Management of Scrotal Mass

Hydrocele vs. Hernia vs. Tumor

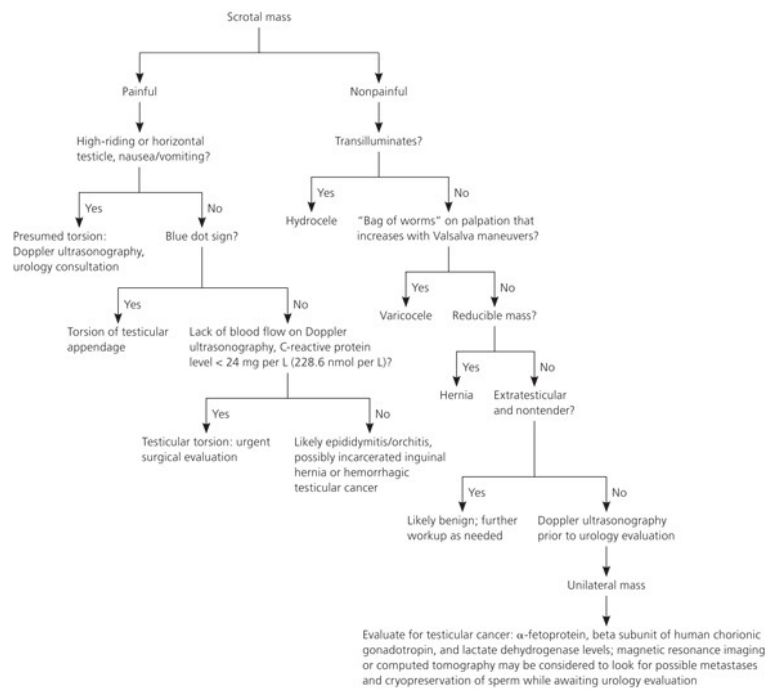
- Investigations

- Ultrasound
- Tumor?:
 - α -fetoprotein, beta hCG, lactate dehydrogenase,
 - CT/MRI to look for metastases
 - Collect sperm for cryopreservation before treatment!



In primary care, discovery of a tender or nontender testicular mass calls for immediate referral to a urologist for optimum care. When you make the referral, you must indicate your concern about testicular cancer so that the urologist knows that this patient cannot wait days or weeks for an appointment!

Fixed, solid, and painless scrotal mass that is attached to the testis must be evaluated for suspected testicular cancer.



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This is a great algorithm for managing a scrotal mass and making decisions about appropriate investigations.

<https://www.aafp.org/afp/2014/0501/p723.html#afp20140501p723-f1>

Table 1. Overview of the Causes of Scrotal Masses

<i>Cause</i>	<i>Clinical presentation</i>	<i>Diagnosis</i>	<i>Treatment</i>
Testicular torsion	Acute unilateral pain and swelling Abnormal cremasteric reflex High position of the testicle Nausea/vomiting See Table 2	Clinical, with or without ultrasonography	Surgery
Epididymitis/orchitis	Acute unilateral pain and swelling Dysuria Erythema of the scrotal skin Fever	Clinical, with or without ultrasonography	Ceftriaxone (Rocephin) and doxycycline
Torsion of the testicular appendage	Acute unilateral pain Blue dot sign (i.e., bluish discoloration of the scrotum over the superior pole)	Ultrasonography	Pain control
Hematocele or testicular rupture	History of trauma Pain and swelling	Ultrasonography or surgical exploration	Pain control Surgery if needed
Testicular cancer	Firm, unilateral nodule	Ultrasonography Tumor markers	Surgery
Inguinal hernias	Pain with Valsalva maneuvers Unilateral bulge in the scrotum	Physical examination Ultrasonography	Surgery
Hydrocele	Swelling	Transillumination Ultrasonography	Pain control Surgery if needed
Varicocele	Dull ache when standing Scrotal mass	"Bag of worms" on palpation	Scrotal support Surgery if needed
Skin cancer	History of carcinogens Erosive, vascular, hyperkeratotic, or nonhealing; color changes; irregular border	Biopsy	Surgery

Information from references 1, 2, 4, and 6 through 8.

Erectile Dysfunction

- **Inability to attain or sustain an erection satisfactory for sexual intercourse.**
- Most erectile dysfunction is related to vascular, neurologic, psychologic, and hormonal disorders; drug use can also be a cause.
- Evaluation typically includes screening for underlying disorders and measuring testosterone levels.
- Treatment options include
 - Oral phosphodiesterase inhibitors, (such as Viagra- Sildenafil)
 - Intraurethral or intracavernosal prostaglandins
 - Vacuum erection devices
 - Surgical implants

Erectile Dysfunction - causes

Organic ED (90% cases):	
1. Vascular ED: due to Hypertension, Diabetes, Smoking etc.	Diminished peripheral pulses suggest vascular dysfunction.
2. Neurogenic ED: due to Diabetic neuropathy, Surgical injury to nerves, Stroke, Seizures, Multiple sclerosis, Spinal cord injuries etc.	Poor rectal tone, decreased perineal sensation, or abnormal bulbocavernosus reflexes may indicate neurologic dysfunction.
3. Hormonal ED: Hypogonadism (testosterone deficiency), hyperprolactinemia, or thyroid dysfunction	Look for associated clinical features
4. Anatomic ED: Peyronie disease, Hypo/epispadias, Phimosis	Genital examination to evaluate
4. Drug/ substance induced: like Beta-blockers, Spironolactone, Alcohol, Cocaine, SSRI, TCA antidepressants; Anticholinergics etc.	Careful drug history
Psychologic (non-organic) ED (10% cases):	
• Abrupt onset with clear precipitating event • Causes may relate to performance anxiety, stress, or depression • May be situational, involving a particular place, time, or partner • Presence of spontaneous nocturnal, morning erections strongly support psychologic ED (as there is no underlying neurovascular or hormonal problems)	

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- ED with normal or intact nocturnal/ early morning erections is strongly suggestive of 'psychogenic ED'

Evaluation and Management of Erectile Dysfunction

- History is all important – see previous slide
- Are co-morbidities present or possible (DM, ASHD, peripheral vascular disease, neurologic disease, etc.)?
- Spontaneous morning erections/tumescence (ASK!) are a sign that the problem may be situational/ emotional/ stress-related
- Nocturnal penile tumescence testing for nocturnal or morning erections can be helpful
- Medication is effective – sildenafil, etc.



Tumescence monitoring devices

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Tumescence is another word for erection.

Since the occurrence of morning erections helps you with the differential diagnosis, this means you **MUST** ask the patient about morning erections!

Thank you